

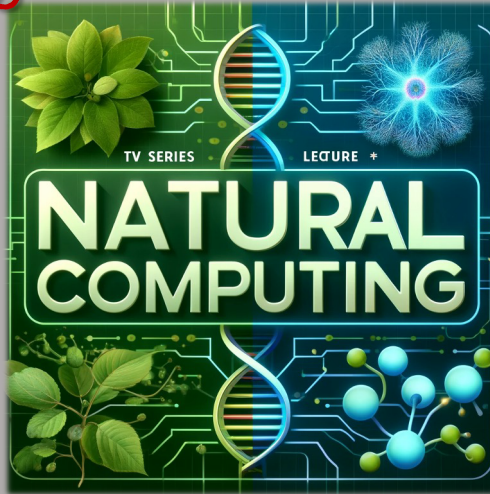
Natural Computing

LMU Munich
summer term 2025

Thomas Gabor

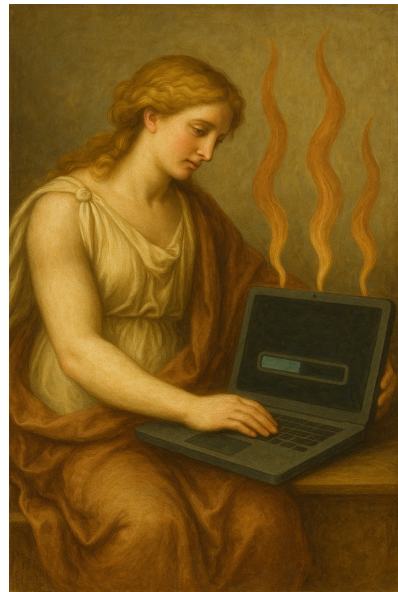


Previously on...



A little excursion:

Why do computers get hot
when they compute?



A little excursion:

all except measuring

Why are most quantum operators
reversible?

Recap

Definition 6 (quantum gate). Let $q = \sum_{i=0}^{2^n-1} c_i |i\rangle$ be an n -qubit register. We often write

$$q = \begin{bmatrix} c_{0\dots 0} \\ \vdots \\ c_{1\dots 1} \end{bmatrix}$$

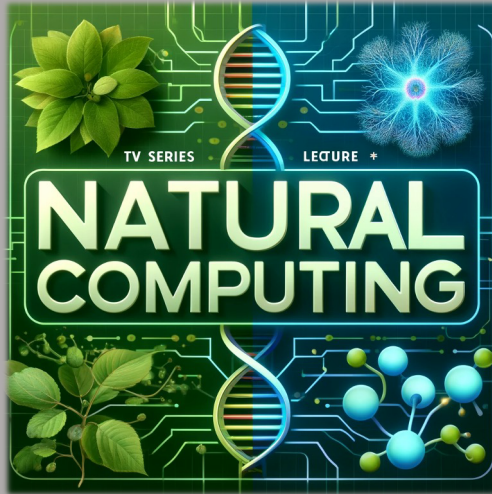
as a shorter notation when the basis $|0\dots 0\rangle, \dots, |1\dots 1\rangle$ is fixed.

An n -ary quantum gate is an operator that works on an n -qubit register q , i.e., (possibly) alters its superposition amplitudes, and thus produces a new n -qubit register q' . An n -ary quantum gate can be given via a matrix $U \in \mathbb{C}^{n \times n}$ so that

$$q' = Uq.$$

Quantum gates commonly implemented on quantum computers are unitary, which — among other things — implies they are reversible.

Operator	Gate(s)	Matrix
Pauli-X (X)	 	
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		
Controlled Z (CZ)	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$



And now for the epic conclusion!

Quantum Advantage

Are there problems for which a quantum algorithm is substantially better than any classical one?

"trivial" case : problem = predicting what a quantum computer would do

One more time!

Theorem 1 (Church-Turing thesis). Any computable function can be computed by a Turing machine.

Theorem 2 (extended Church-Turing thesis). Any computable function can be computed by a Turing machine with at most polynomial overhead.

Quantum Advantage?

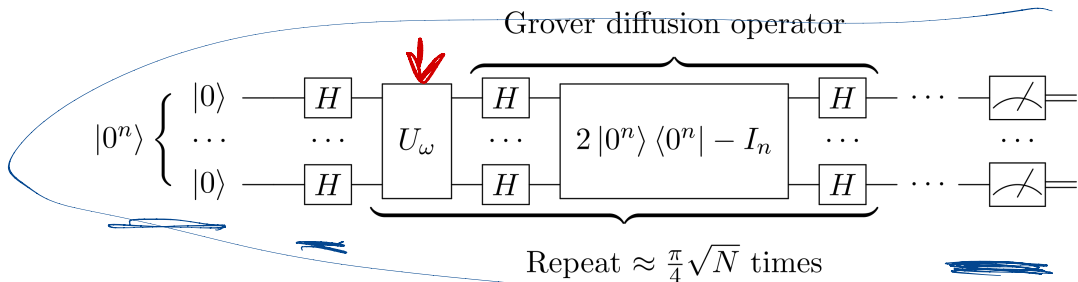
Grover's Search

Problem: Search in unsorted data structure with N elements

Classical Complexity: $\mathcal{O}(N)$

Quantum Complexity: $\mathcal{O}(\sqrt{N})$

en.wikipedia.org/wiki/Grover%27s_algorithm



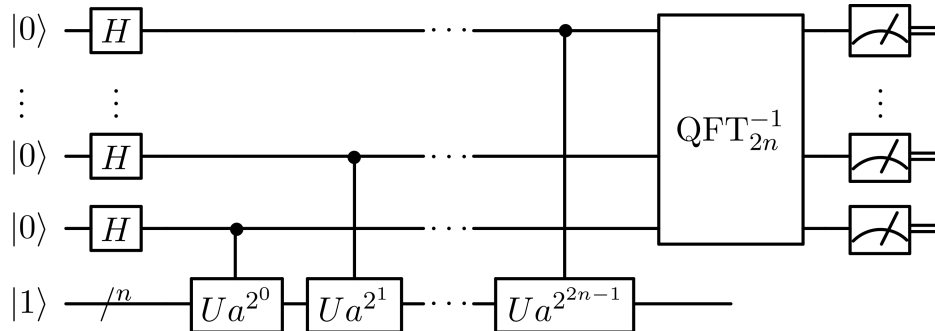
Shor's Algorithm

Problem: Prime factorization of integer N

Classical Complexity: best-known exponential in $\log N$

Quantum Complexity: polynomial in $\log N$

en.wikipedia.org/wiki/Shor%27s_algorithm



HHL Algorithm

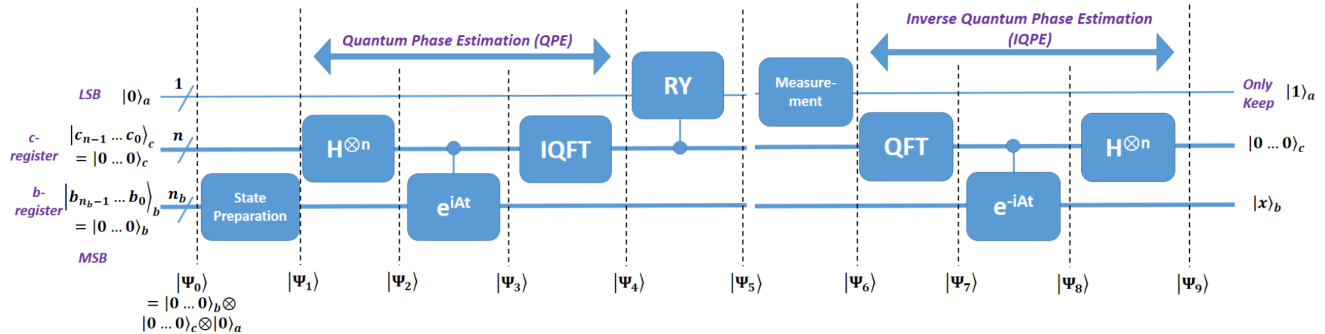
named after
Harrow, Hassidim, Lloyd

Problem: Solving linear systems of N equations with condition number κ

Classical Complexity: $\mathcal{O}(\sqrt{\kappa}N)$ for a scalar value computed from solution

Quantum Complexity: $\mathcal{O}(\kappa^2 \log N)$ for a scalar value computed from solution

adapted from arxiv.org/pdf/2108.09004.pdf



Quantum Annealing

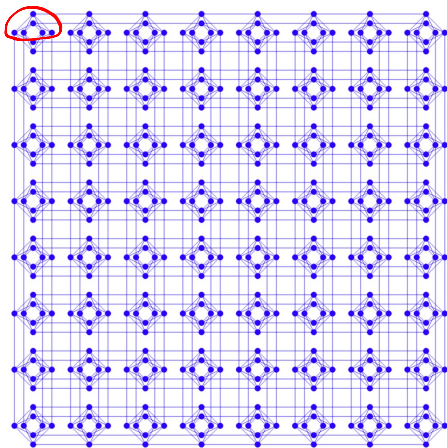
introduced by
Kadowaki, Nishimori

Problem: Solving (quadratic unconstrained binary) optimization problems

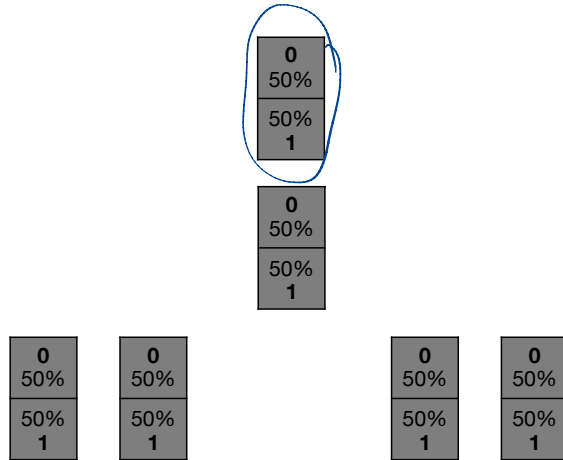
Classical Complexity: best-known exponential for exact solution

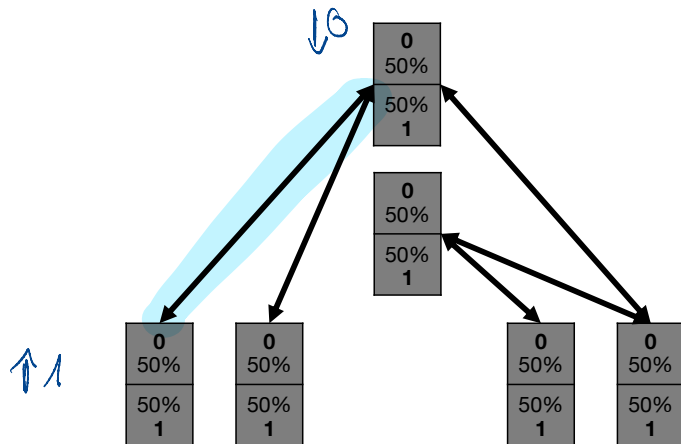
Quantum Complexity: probably exponential as well?

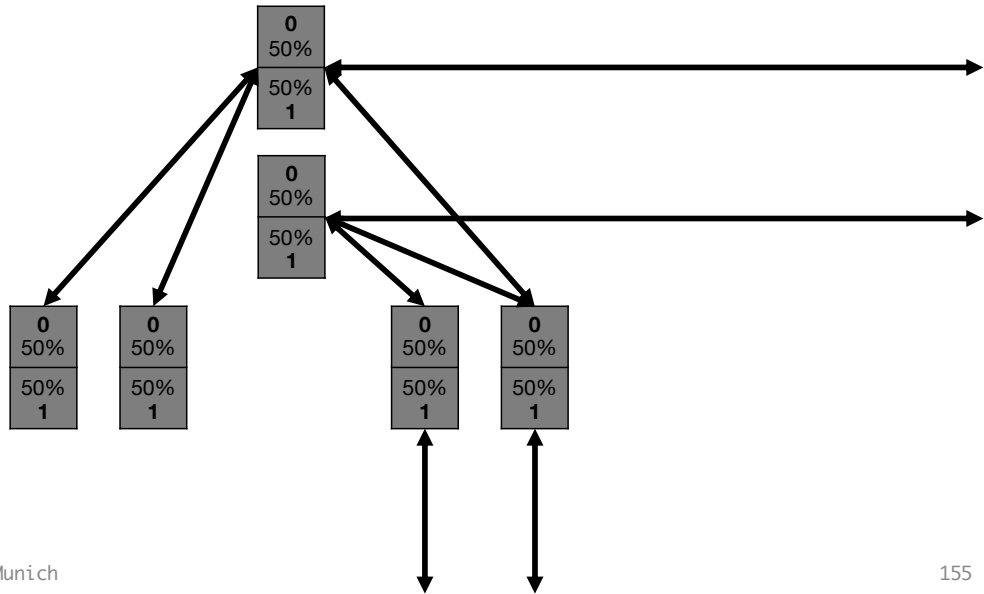
arxiv.org/pdf/1406.2741.pdf

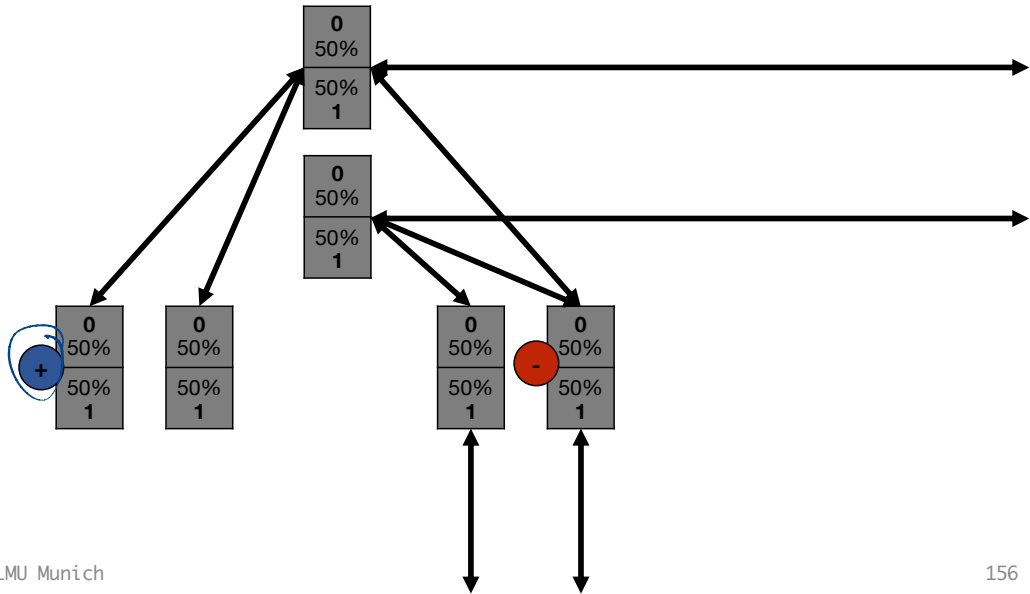


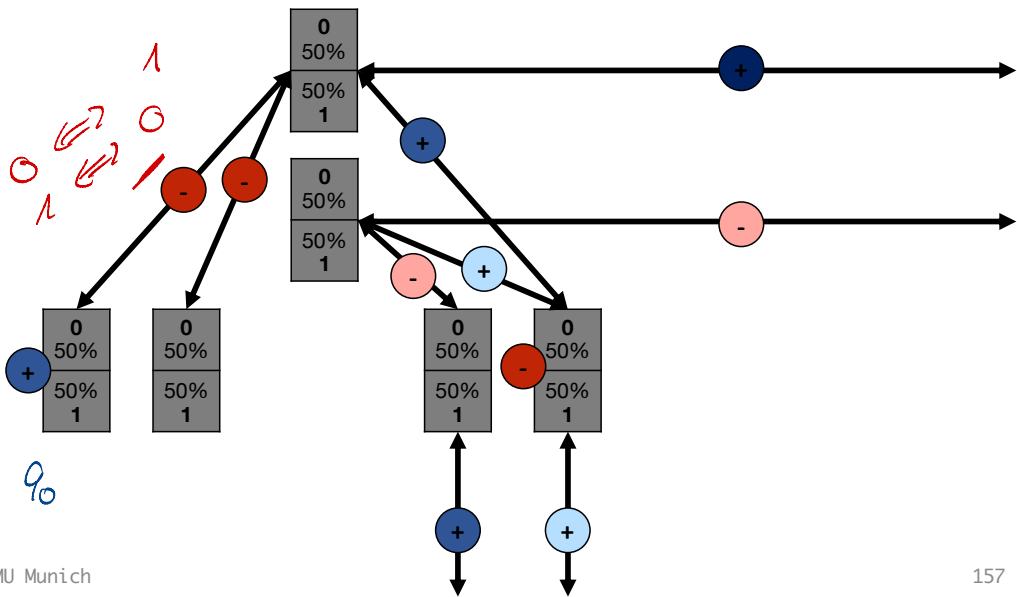
So what is
quantum annealing?





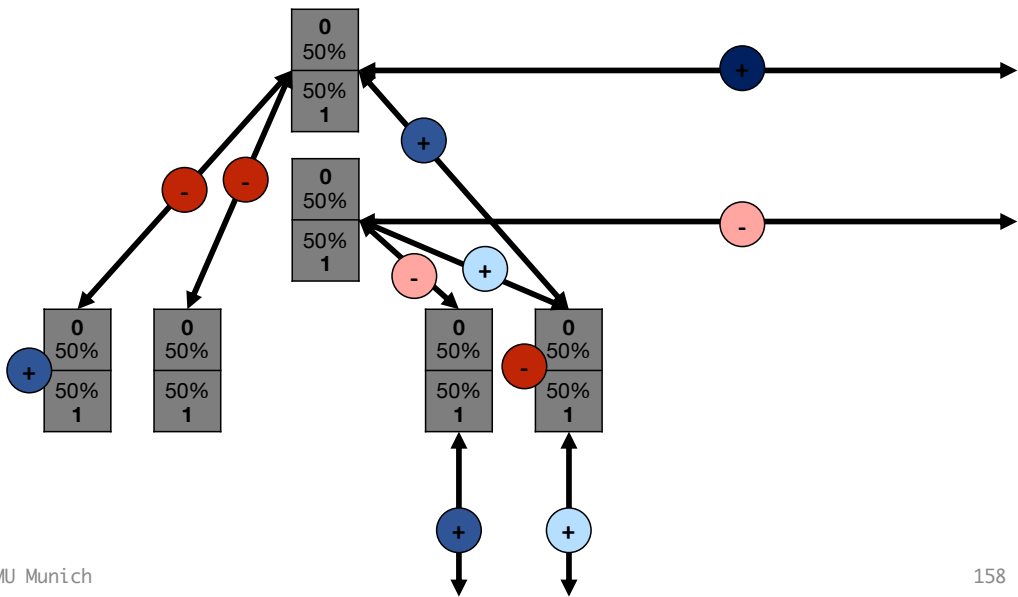




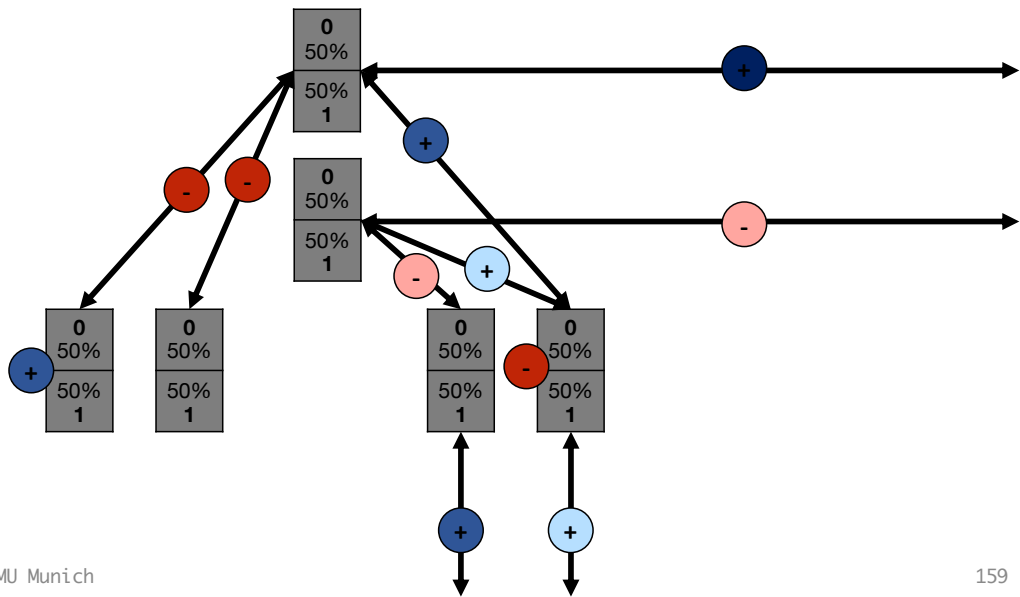


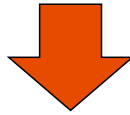
adiabatic quantum computing

add constraints
infinitely slowly
while adding
no energy

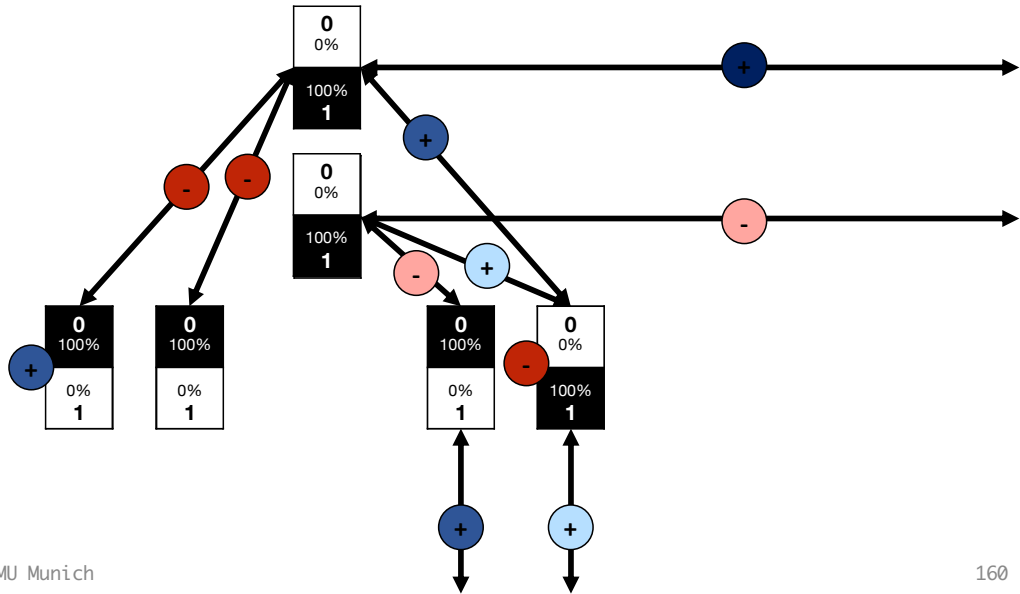


practical quantum annealing





measuring



Let's take a step back!

Experiment Time!

Experiment Time!

